1. Review of Riemann Integration

REAL VARIABLES I

Fall 2026

We briefly review the construction and properties of the Riemann integral and identify its shortcomings that suggest constructing a new integration theory: the Lebesgue integral.

First, we show that continuous functions are a natural class on which the Riemann integral is defined. Some discontinuities can also be tolerated. The exact criterion, proved later in the course, is expressed through the measure-theoretic size of the discontinuity set.

Second, we use the integral to measure the distance between functions and thereby introduce the L^1 norm. We will show that the continuous functions are not complete in the resulting distance, much as the rational numbers are not complete in the usual distance.

Abstract metric-space theory supplies a unique *completion* of the space of continuous functions, and the integral extends continuously to that completion. Lebesgue theory will identify the completed elements concretely as equivalence classes of measurable functions. This gives us an important goal: construct an integration theory whose associated spaces of integrable functions are complete.

After studying these notes, you will be able to do the following.

- Use upper and lower sums to verify Riemann integrability.
- Explain how uniform continuity controls the gap between Riemann sums.
- Construct Cauchy sequences that expose the incompleteness of continuous and Riemann-integrable functions in the integral distance.
- Explain how metric completion motivates the construction of the Lebesgue integral.

1.1 Riemann integrability

A function is Riemann integrable if lower and upper Riemann sum approximations can be made arbitrarily close.

Definition 1.1.1 (Partition of an interval)**.** For $a, b \in \mathbb{R}$ with $a < b$, a **partition** of $[a, b]$ is a finite ordered tuple $\tau = (\tau_0, \tau_1, \dots, \tau_n)$ such that $a = \tau_0 < \tau_1 < \cdots < \tau_n = b$. The **mesh** of the partition is

$$|\tau| \coloneqq \max_{0 \le j \le n-1} (\tau_{j+1} - \tau_j).$$

The number of intervals in the partition is denoted by $\#\tau \coloneqq n$.

Warning (Two meanings of partition)**.** The term *partition* is heavily overloaded in mathematics. Often it stands for an arbitrary representation of a set as a disjoint union. Here, we partition an interval into *finitely many* ordered intervals.

Definition 1.1.2 (Upper and lower Riemann sums)**.** For a bounded function $f\colon [a, b] \to \mathbb{R}$ and a partition

τ, the **upper Riemann sum** and **lower Riemann sum** are

$$U(f;\tau) \coloneqq \sum_{j=0}^{\#\tau-1} \Big(\sup_{x\in[\tau_j,\tau_{j+1}]} f(x)\Big)(\tau_{j+1}-\tau_j)$$
$$L(f;\tau) \coloneqq \sum_{j=0}^{\#\tau-1} \Big(\inf_{x\in[\tau_j,\tau_{j+1}]} f(x)\Big)(\tau_{j+1}-\tau_j).$$

The lower sum never exceeds the upper sum. Refining a partition can only raise the lower sum and lower the upper sum.

Definition 1.1.3 (Riemann integrability)**.** A bounded function $f\colon [a,b] \to \mathbb{R}$ is **Riemann integrable** if

$$\sup_{\tau} L(f;\tau) = \inf_{\tau} U(f;\tau),$$

where the supremum and infimum are taken over all finite partitions of $[a,b]$.
The common value is denoted by

$$\int_a^b f(x)\,\mathrm{d}x.$$

The Darboux criterion says that f is Riemann integrable if and only if, for every $\varepsilon > 0$, some partition τ satisfies $U(f;\tau) - L(f;\tau) < \varepsilon$. For the nontrivial direction, choose partitions whose upper and lower sums lie within $\varepsilon/2$ of the common integral value. A common refinement raises the lower sum and lowers the upper sum, producing the required gap.
According to this reduction, we must show that

$$U(f;\tau) - L(f;\tau) = \sum_{j=0}^{\#\tau-1} \operatorname{osc}(f;[\tau_j,\tau_{j+1}])\,(\tau_{j+1}-\tau_j), \tag{1}$$

can be made arbitrarily small, where the oscillation of f over an interval J is defined as

$$\operatorname{osc}(f;J) \coloneqq \sup_{x\in J} f(x) - \inf_{x\in J} f(x).$$

We are ready to prove the following theorem.

Theorem 1.1.4 (Continuous functions are Riemann integrable)**.** Every continuous function $f\colon [a,b] \to \mathbb{R}$ is Riemann integrable.

Proof. (proves 1.1.4)
The proof uses uniform continuity to control every oscillation in (1) with one scale. Fix $\varepsilon > 0$. By the Heine–Cantor theorem, there is $\delta > 0$ such that

$$|x-y| < \delta \quad \Longrightarrow \quad |f(x)-f(y)| < \frac{\varepsilon}{b-a} \qquad (x,y \in [a,b]).$$

Choose a partition τ with $|\tau| < \delta$. The maximum and minimum of f on each partition interval are attained, so

$$\operatorname{osc}(f;[\tau_j,\tau_{j+1}]) < \frac{\varepsilon}{b-a}.$$

Consequently,

$$U(f;\tau) - L(f;\tau) < \frac{\varepsilon}{b-a} \sum_{j=0}^{\#\tau-1} (\tau_{j+1} - \tau_j) = \varepsilon.$$

The Darboux criterion proves that f is Riemann integrable. $\square$

1.1.1 Uniform continuity and moduli of continuity

The proof above essentially relies on the following reasoning.
Suppose every subinterval of a partition has length at most δ. If the values of f differ by at most η whenever their inputs are within δ of one another, then every oscillation in (1) is at most η, and therefore

$$U(f;\tau) - L(f;\tau) \leq \eta(b-a).$$

The point is that one value of δ must work everywhere on $[a,b]$, not merely near one point at a time.

Definition 1.1.5 (Uniform continuity)**.** For $E \subseteq \mathbb{R}$, a function $f\colon E \to \mathbb{R}$ is **uniformly continuous** on E if, for every $\varepsilon > 0$, there exists $\delta > 0$ such that for all $x, y \in E$,

$$|x-y| < \delta \quad \Longrightarrow \quad |f(x) - f(y)| < \varepsilon.$$

The number δ may depend on ε, but it must work for every pair $x, y \in E$.

For continuity at a point x_0, the choice of δ may depend on x_0. A function can therefore be continuous at every point while requiring smaller and smaller scales as the point moves through the domain. Uniform continuity rules this out by requiring one common scale.

Theorem (Review: Heine–Cantor theorem)**.** For a compact set $K \subseteq \mathbb{R}$, every continuous function $f\colon K \to \mathbb{R}$ is uniformly continuous. In particular, every continuous function on a closed bounded interval $[a,b]$ is uniformly continuous.

The familiar proof extracts a finite subcover from neighborhoods on which the continuity estimates hold.
We use the theorem here as a review result.
Generally, continuous functions are not uniformly continuous.

Problem (Uniform continuity of power functions)**.** For $\alpha > 0$, prove that $x \mapsto |x|^\alpha$ is continuous on $\mathbb{R}$ and is uniformly continuous there if and only if $\alpha \leq 1$.

The modulus of continuity encodes the $\varepsilon - \delta$ relationship of Definition 1.1.5 by recording how much a function can change at each spatial scale.

Definition (Modulus of continuity)**.** For an interval $I \subseteq \mathbb{R}$ and a function $f\colon I \to \mathbb{R}$, a **modulus of continuity** for f is a nondecreasing function

$$\omega : [0,\infty) \to [0,\infty)$$

such that

$$\omega(t) \longrightarrow 0 \qquad \text{as } t \to 0^+,$$

and

$$|f(x_2) - f(x_1)| \leq \omega(|x_2 - x_1|) \qquad \forall x_1, x_2 \in I.$$

The natural choice is the largest change of value of f over all pairs separated by at most δ:

$$\omega_f(\delta) \coloneqq \sup\{|f(x_2) - f(x_1)| \,:\, x_1, x_2 \in I,\ |x_2 - x_1| \le \delta\}. \tag{2}$$

This definition packages the estimate used in the preceding proof.

Proposition (Review: Uniform continuity is equivalent to having a modulus)**.** For an interval or a finite union of intervals $E \subseteq \mathbb{R}$, a function $f\colon E \to \mathbb{R}$ is uniformly continuous if and only if it has a modulus of continuity.
When f is uniformly continuous, the function ω_f in (2) is finite for every $\delta > 0$ and is a modulus of continuity.

The modulus gives a quantitative form of the uniform estimate and will also separate the regular and exceptional intervals in the next proof.

1.1.2 Riemann integrability of functions with discontinuities

Continuity controlled the oscillation on every subinterval. If a bounded function has finitely many discontinuities, that control fails near those points. The proof treats two regions differently: continuity makes the oscillation small away from the discontinuities, while boundedness combines with the short total length of the exceptional intervals near them.

Proposition 1.1.6 (Bounded functions with finitely many discontinuities are Riemann integrable)**.** For a bounded function $f\colon [a,b] \to \mathbb{R}$, finitely many discontinuities imply Riemann integrability.

Proof. (proves 1.1.6)
The proof controls regular intervals by uniform continuity and exceptional intervals by their total length. Let $D = \{d_1, \dots, d_N\}$ be the set of discontinuities and put $M \coloneqq \|f\|_{\sup} < \infty$.

Step 1: isolate the discontinuities.

For $r > 0$, define the exceptional set

$$E_r \coloneqq [a,b] \cap \bigcup_{k=1}^{N} (d_k - r, d_k + r).$$

The complement $[a,b] \setminus E_r$ is a finite union of compact intervals on which f is continuous. Hence f has a modulus of continuity ω_r on this complement.
For a partition τ, let

$$\mathcal{J}_{\tau,r} \coloneqq \left\{ j \in \{0, \dots, \#\tau - 1\} \,:\, [\tau_j, \tau_{j+1}] \cap E_r \neq \emptyset \right\}.$$

These are precisely the indices of the partition intervals that meet the exceptional set.

Step 2: estimate the two kinds of intervals.

For any partition τ,

$$\begin{aligned} U(f;\tau) - L(f;\tau) = &\sum_{j \notin \mathcal{J}_{\tau,r}} \operatorname{osc}(f; [\tau_j, \tau_{j+1}])\,(\tau_{j+1} - \tau_j) \\ &+ \sum_{j \in \mathcal{J}_{\tau,r}} \operatorname{osc}(f; [\tau_j, \tau_{j+1}])\,(\tau_{j+1} - \tau_j) \end{aligned}$$

The regular intervals lie in $[a,b] \setminus E_r$, so

$$\sum_{j \notin \mathcal{J}_{\tau,r}} \operatorname{osc}(f; [\tau_j, \tau_{j+1}])\,(\tau_{j+1} - \tau_j) \le \omega_r(|\tau|)(b-a).$$

On every exceptional interval the oscillation is at most $2M$. Moreover,

$$\bigcup_{j\in\mathcal{J}_{\tau,r}} [\tau_j, \tau_{j+1}] \subseteq E_{r+|\tau|},$$

because each such interval has length at most $|\tau|$ and meets an r-neighborhood of a point in D. Therefore

$$\sum_{j\in\mathcal{J}_{\tau,r}} (\tau_{j+1} - \tau_j) \leq 2N(r + |\tau|),$$

and hence

$$\sum_{j\in\mathcal{J}_{\tau,r}} \operatorname{osc}(f; [\tau_j, \tau_{j+1}])\,(\tau_{j+1} - \tau_j) \leq 4MN(r + |\tau|).$$

Step 3: make the Darboux gap vanish.

Combining the estimates gives

$$U(f;\tau) - L(f;\tau) \leq \omega_r(|\tau|)(b-a) + 4MN(r + |\tau|).$$

Letting the mesh tend to zero gives

$$\inf_{\tau}(U(f;\tau) - L(f;\tau)) \leq 4MNr.$$

Since $r > 0$ is arbitrary,

$$\inf_{\tau}(U(f;\tau) - L(f;\tau)) = 0.$$

The Darboux criterion proves the proposition. □

This proof shows that finiteness is not the essential issue. What mattered was the ability to cover all discontinuities by intervals whose total length is as small as desired. That observation leads to the exact characterization that will be proved later in the course.

Definition 1.1.7 (Lebesgue null set)**.** A set $E \subseteq \mathbb{R}$ has **Lebesgue measure zero**, or is a **Lebesgue null set**, if for every $\varepsilon > 0$ there exists a sequence of open intervals $(a_k, b_k)_{k\in\mathbb{N}}$ such that

$$E \subseteq \bigcup_{k=1}^{\infty} (a_k, b_k)$$

and

$$\sum_{k=1}^{\infty} |b_k - a_k| < \varepsilon.$$

A null set need not be finite. The definition says that, from the point of view of total interval length, the set can be confined to a region of arbitrarily small "length."

Theorem 1.1.8 (Preview: Lebesgue's criterion for Riemann integrability)**.** For a bounded function $f\colon [a,b] \to \mathbb{R}$, define

$$D_f \coloneqq \{x \in [a,b] \,:\, f \text{ is discontinuous at } x\}.$$

Then f is Riemann integrable if and only if D_f has Lebesgue measure zero.

No proof is given here. The theorem states the destination of the first part of the course: Riemann integrability is governed not by the number of discontinuities, but by whether their locations can be covered by intervals of arbitrarily small total length. This last quantity is a prototypical construction in Lebesgue measure theory.

1.2 The L^1 norm and incompleteness

On $C([a,b])$, define

$$\|f\|_{L^1} \coloneqq \int_a^b |f(x)|\,\mathrm{d}x,$$

This quantity measures the integral magnitude of f; dividing by $b-a$ would give its average magnitude. It defines the distance

$$d_1(f,g) \coloneqq \|f-g\|_1.$$

The quantity $d_1(f,g)$ measures the integral discrepancy between f and g. A large pointwise error can have small integral cost when it is confined to a short interval.

Before testing limits, we verify that $\|\cdot\|_1$ is a norm on continuous functions. Nonnegativity, homogeneity, and the triangle inequality follow from the corresponding properties of the absolute value. The remaining issue is whether zero integral forces a continuous function to vanish everywhere.

Lemma 1.2.1 (Zero integral forces a continuous function to vanish). Let $f \in C([a,b])$. If

$$\int_a^b |f(x)|\,\mathrm{d}x = 0,$$

then $f = 0$ on $[a,b]$.

Proof. (proves 1.2.1)
The proof turns a nonzero value into a positive lower bound on an interval. Suppose that $f(x_0) \neq 0$ for some $x_0 \in [a,b]$. By continuity at x_0, there exists $r > 0$ such that

$$|f(x)| \geq \frac{|f(x_0)|}{2}$$

for every

$$x \in [a,b] \cap (x_0 - r, x_0 + r).$$

This set contains an interval of positive length. Integrating the positive lower bound over that interval gives

$$\int_a^b |f(x)|\,\mathrm{d}x > 0,$$

contrary to the hypothesis. Hence f vanishes everywhere. $\square$

We can now ask whether every sequence that should have a limit in this distance actually has one in $C([a,b])$.

Definition 1.2.2 (Complete metric space). A sequence (f_n) in a metric space is **Cauchy** if, for every $\varepsilon > 0$, there exists N such that

$$d(f_m, f_n) < \varepsilon \qquad \forall m,n \geq N.$$

A metric space is **complete** if every Cauchy sequence converges (i.e. has a limit).

However, $C^0([a,b])$ is not complete with respect to the L^1 distance.

Proposition 1.2.3 (Continuous functions are not complete in the integral norm)**.** The normed space

$$(C([-1,1]), \|\cdot\|_{L^1}), \qquad \|f\|_{L^1} \coloneqq \int_{-1}^{1} |f(x)| \, \mathrm{d}x,$$

is not complete.

Proof. (proves 1.2.3)
The sequence below sharpens a jump across intervals whose lengths tend to zero. We first verify that it is Cauchy and then show that continuity excludes any possible limit.
For $n \in \mathbb{N}$, define

$$f_n(x) = \begin{cases} 0, & -1 \le x \le 0, \\ 2^n x, & 0 < x < 2^{-n}, \\ 1, & 2^{-n} \le x \le 1. \end{cases}$$

Each f_n is continuous.

Step 1: verify the Cauchy property.

If $m \ge n$, then

$$\|f_m - f_n\|_{L^1} \le 2^{-n}.$$

Indeed, $|f_m(x) - f_n(x)| \le \mathbb{1}_{[0,2^{-n}]}(x)$.

Step 2: rule out a continuous limit.

Suppose for contradiction that $f_n \to f$ in L^1 for some $f \in C([-1,1])$. We first determine the value of $f(0)$.
If $f(0) \ne 0$, continuity at 0 gives $\delta > 0$ such that $|f(x)| \ge \frac{|f(0)|}{2}$ for every $x \in [-\delta, 0]$. For every n, we have $f_n(x) = 0$ on $[-1, 0]$. Consequently,

$$\|f_n - f\|_{L^1} \ge \int_{-\delta}^{0} |f_n(x) - f(x)| \, \mathrm{d}x = \int_{-\delta}^{0} |f(x)| \, \mathrm{d}x \ge \delta \frac{|f(0)|}{2} > 0.$$

This contradicts $\|f_n - f\|_{L^1} \to 0$ and thus, necessarily, $f(0) = 0$.
We now use the behavior to the right of 0. Since f is continuous at 0 and $f(0) = 0$, there exists $\delta > 0$ such that $|f(x)| < \frac{1}{2}$ for every $x \in [0, \delta]$. Choose n sufficiently large that $2^{-n} < \frac{\delta}{2}$. For every $x \in [2^{-n}, \delta]$, we have $f_n(x) = 1$, and therefore

$$|f_n(x) - f(x)| = |1 - f(x)| \ge 1 - |f(x)| > \frac{1}{2}.$$

It follows that

$$\|f_n - f\|_{L^1} \ge \int_{2^{-n}}^{\delta} |f_n(x) - f(x)| \, \mathrm{d}x > \frac{1}{2}(\delta - 2^{-n}) > \frac{\delta}{4}.$$

This lower bound holds for every sufficiently large n, contradicting the assumption that $\|f_n - f\|_{L^1} \longrightarrow 0$. Therefore (f_n) has no limit in $C([-1,1])$ with respect to the L^1-norm. Hence $(C([-1,1]), \|\cdot\|_{L^1})$ is not complete. □

One can guess what the missing limit function should be: it is the step function $h = \mathbb{1}_{[0,1]}$. The sequence f_n fails to converge inside $C([-1,1])$ because continuity is not preserved by convergence in L^1.
The step function is Riemann integrable, so one might try to repair the problem by replacing continuous functions with all Riemann-integrable functions.
A more subtle argument is required. The fat-rationals construction later in these notes will show directly

that Riemann-integrable functions are not complete with respect to the L^1 distance.

Warning (The integral quantity is only a seminorm on Riemann-integrable functions). Let $\mathcal{R}([0,1])$ be the vector space of Riemann-integrable functions on $[0,1]$.
For

$$f(x) = \begin{cases} 1, & x = 0, \\ 0, & x \neq 0, \end{cases}$$

we have $f \neq 0$ but $\|f\|_1 = 0$.
Thus $\|\cdot\|_1$ is not a norm on pointwise-defined Riemann-integrable functions. To obtain a norm, functions at zero distance must be identified.

Even if we carefully avoid the problem above, some Cauchy sequences of Riemann-integrable functions still fail to converge to a Riemann-integrable limit function.

Proposition (Riemann-integrable functions are incomplete with respect to L^1). There exists a sequence $(f_n)_{n\in\mathbb{N}}$ of Riemann-integrable functions on $[0,1]$ that is Cauchy with respect to

$$d_1(f,g) = \int_0^1 |f(x) - g(x)| \, \mathrm{d}x$$

but there does not exist a Riemann-integrable function h on $[0,1]$ such that $\lim_n d_1(f_n, h) = 0$.

The proof is deferred until after the explicit fat-rationals construction below; it uses only Riemann integration.

1.3 The abstract repair: completion

The examples tell us what is wrong with the starting spaces: the integral distance produces Cauchy sequences whose limits are absent. We can use abstract metric completion to "fill in the gaps" and add one new element for each missing limiting behavior.

Definition (Completion of a metric space). A **completion** of a metric space (X,d) is a complete metric space $(\widehat{X}, \hat{d})$ together with an isometric embedding

$$\iota \colon X \to \widehat{X}$$

such that $\iota(X)$ is dense in $\widehat{X}$.

Theorem (Review: Existence and uniqueness of metric completion). Every metric space has a completion. Any two completions are isometric by an isometry that preserves the embedded copy of the original space.

In the standard construction, an element of $\widehat{X}$ is an equivalence class of Cauchy sequences in X: two Cauchy sequences represent the same element when the distance between their corresponding terms tends to zero. Applied to $C([a,b])$ with the integral norm, this construction formally supplies all missing limits. The integral itself extends to the completion by the next review result.

Proposition (Review: Uniformly continuous maps extend to completions). Let (X, d_X) be a metric space, let $\widehat{X}$ be a completion of X, and let (Y, d_Y) be complete. Every uniformly continuous map

$$T \colon X \to Y$$

has a unique continuous extension

$$\hat{T}\colon \widehat{X} \to Y.$$

Define the integration map as

$$I\colon C([a,b]) \to \mathbb{R}, \qquad I(f) \coloneqq \int_a^b f(x)\ \mathrm{d}x.$$

For $f, g \in C([a,b])$,

$$|I(f) - I(g)| = \left|\int_a^b (f-g)(x)\ \mathrm{d}x\right| \le \int_a^b |f(x) - g(x)|\ \mathrm{d}x = \|f-g\|_1.$$

Thus the integral is 1-Lipschitz: functions that are close in integral norm have close integrals. Because I is Lipschitz and $\mathbb{R}$ is complete, I extends uniquely to the completion of $C([a,b])$ in the integral norm. Concretely, if an element of the completion is represented by a Cauchy sequence (f_n), its integral is defined by

$$\hat{I}([(f_n)]) \coloneqq \lim_{n\to\infty} \int_a^b f_n(x)\ \mathrm{d}x.$$

The Lipschitz estimate guarantees both that this limit exists and that equivalent Cauchy sequences give the same value.

This construction accomplishes two goals: it creates a complete space and extends the integral to every new element. It does not yet give a satisfactory description of those elements. Equivalence of L^1 Cauchy sequences of continuous functions is an abstract object; that description alone does not determine whether the new element is representable as a ”function.”

The measure-theoretic construction provides a concrete model for this abstract completion. After Lebesgue measure and integration have been developed, the completion above can be identified with the *Lebesgue* space $L^1([a,b])$.

1.4 Open dense sets of small total length: the fat rationals

We recall two elementary topological notions.

Definition (Open and dense subsets)**.** For a metric space (X,d), a set $E \subseteq X$ is **open** if for every $x \in E$ there exists $r_x > 0$ such that $B_{r_x}(x) \subseteq E$.
The set $E \subset X$ is **dense** if any one of the following equivalent conditions holds:

- its closure $\bar{E}$ is X;
- every nonempty open subset of X intersects E.

We henceforth reason on the metric space $[0,1]$ with the standard Euclidean distance. A set can be both open and dense. For example, the sets $(0,1)$ or $[0,1/2) \cup (1/2,1]$ are open and dense in $[0,1]$.

We now give a different construction that also controls the total length of the intervals used to form the open set. List all rational numbers in $[0,1]$, allowing repetitions, by successively listing the fractions with each positive denominator:

$$\begin{aligned}
&q_0 = \frac{0}{1}, \qquad q_1 = \frac{1}{1},\\
&q_2 = \frac{0}{2}, \qquad q_3 = \frac{1}{2}, \qquad q_4 = \frac{2}{2},\\
&q_5 = \frac{0}{3}, \qquad q_6 = \frac{1}{3}, \qquad q_7 = \frac{2}{3}, \qquad q_8 = \frac{3}{3}, \qquad \ldots
\end{aligned}$$

Every rational number in $[0,1]$ appears somewhere in this sequence. Thus $\mathbb{Q} \cap [0,1] = \{q_n : n \geq 0\}$.

Definition 1.4.1 (Fat rationals in the unit interval). Recall that on $[0,1]$ we have $B_\rho(x) \coloneqq (x-\rho, x+\rho) \cap [0,1]$. Fix $r > 0$ and define

$$F_r \coloneqq [0,1] \cap \bigcup_{n=0}^{\infty} B_{r2^{-n}}(q_n).$$

We refer to F_r informally as the set of *fat rationals*.

Each rational q_n has been replaced by an interval whose radius decreases with n. The set F_r is open in $[0,1]$ since it is a union of open sets. It contains every rational point of $[0,1]$, since $q_n \in B_{r2^{-n}}(q_n)$ for every n. Because the rational numbers are dense, F_r is dense in $[0,1]$.

Intuitively, the intervals in the definition of F_r have total length

$$\sum_{n=0}^{\infty} 2r2^{-n} = 4r.$$

Therefore, when r is sufficiently small, these intervals cannot cover all of $[0,1]$. Next, we make this statement precise.

Proposition 1.4.2 (The fat rationals form a proper subset). If $0 < r \leq 1/4$, then F_r is a proper subset of $[0,1]$.

Proof. (proves 1.4.2)

Compactness reduces the countable cover to a finite one. A chain of overlapping intervals then compares the length of $[0,1]$ with the total length available in the cover.

Step 1: reduce to a finite cover.

Suppose, to the contrary, that $[0,1] = \bigcup_{n=0}^{\infty} B_{r2^{-n}}(q_n)$, i.e. $\left\{B_{r2^{-n}}(q_n)\right\}_{n\in\mathbb{N}}$ is an open cover of the compact interval $[0,1]$. By compactness, there exists a finite set of indices $\mathcal{J}$ such that

$$[0,1] \subseteq \bigcup_{n\in\mathcal{J}} B_{r2^{-n}}(q_n).$$

For $n \in \mathcal{J}$, set

$$a_n \coloneqq q_n - r2^{-n}, \qquad b_n \coloneqq q_n + r2^{-n}.$$

Then $B_{r2^{-n}}(q_n) = (a_n, b_n) \cap [0,1]$.

Step 2: extract an overlapping chain.

Next, we extract a chain of overlapping intervals from the finite cover. Since 0 is covered, there exists $n_0 \in \mathcal{J}$ such that $a_{n_0} < 0 < b_{n_0}$. If $b_{n_0} > 1$, the chain already covers the interval from a point to the left of 0 to a point to the right of 1.

Otherwise, $b_{n_0} \in [0,1]$. Since b_{n_0} is covered by the finite collection of open intervals, there exists $n_1 \in \mathcal{J} \setminus \{n_0\}$ such that $a_{n_1} < b_{n_0} < b_{n_1}$. In particular, $b_{n_1} > b_{n_0}$. If $b_{n_1} \leq 1$, we repeat the argument. At the k-th step, choose an index $n_{k+1} \in \mathcal{J} \setminus \{n_0, \dots, n_k\}$ such that

$$a_{n_{k+1}} < b_{n_k} < b_{n_{k+1}}.$$

The indices chosen at successive steps are distinct by construction, and since $\mathcal{J}$ is finite, the process terminates. We therefore obtain indices $n_0, n_1, \dots, n_M \in \mathcal{J}$ such that

$$a_{n_0} < 0, \qquad a_{n_k} < b_{n_{k-1}} < b_{n_k} \quad \text{for } 1 \leq k \leq M, \qquad b_{n_M} > 1.$$

It follows that

$$1 < b_{n_M} - a_{n_0}.$$

Step 3: compare the available lengths.

The overlap relations give

$$\begin{aligned} b_{n_M} - a_{n_0} &= (b_{n_0} - a_{n_0}) + \sum_{k=1}^{M} (b_{n_k} - b_{n_{k-1}}) \\ &< (b_{n_0} - a_{n_0}) + \sum_{k=1}^{M} (b_{n_k} - a_{n_k}) \\ &= \sum_{k=0}^{M} (b_{n_k} - a_{n_k}) \\ &\le \sum_{n \in \mathcal{J}} (b_n - a_n) \\ &\le \sum_{n=0}^{\infty} 2r2^{-n} = 4r. \end{aligned}$$

Consequently $1 < 4r$, which is a contradiction to our assumption $r \le 1/4$. Therefore $F_r \neq [0, 1]$. □

Problem (The role of compactness in the fat-rationals construction). Where are compactness and finiteness used in the proof of Proposition 1.4.2? Why can the overlapping-chain argument not be applied directly to $\{B_{r2^{-n}}(q_n)\}_{n \in \mathbb{N}}$?

1.4.1 Incompleteness of Riemann integrable functions

The fat-rationals construction now proves Section 1.2. The proof first identifies the natural pointwise limit and then rules out every possible Riemann-integrable limit in the integral distance.

Proof. (proves 1.2)
Fix $0 < r < 1/4$, and define

$$F_n \coloneqq \bigcup_{k=0}^{n} B_{r2^{-k}}(q_k), \qquad F \coloneqq \bigcup_{n \in \mathbb{N}} F_n, \qquad f_n \coloneqq \mathbb{1}_{F_n}.$$

Step 1: verify the Cauchy property.

Each F_n is a finite union of intervals, so f_n has only finitely many discontinuities and is Riemann integrable. If $m \ge n$, then

$$\|f_m - f_n\|_{L^1} = \int_0^1 |f_m(x) - f_n(x)| \, \mathrm{d}x \le \sum_{k=n+1}^{m} 2r2^{-k} \le 2r2^{-n}.$$

Indeed, the sets F_n are increasing and

$$0 \le f_m - f_n \le \sum_{k=n+1}^{m} \mathbb{1}_{B_{r2^{-k}}(q_k) \cap [0,1]}.$$

This shows that (f_n) is Cauchy with respect to d_1.

Step 2: identify the failure of the pointwise limit.

The pointwise limit is $\mathbb{1}_F$, but this function is not Riemann integrable.
Since F contains $\mathbb{Q} \cap [0,1]$, it is dense in $[0,1]$. Hence every nondegenerate subinterval of $[0,1]$ intersects F, and therefore $U(\mathbb{1}_F;\tau) = 1$ for every partition τ. It remains to bound the lower sums. Let

$$\tau = (\tau_0, \dots, \tau_{\#\tau})$$

be a partition, and let $\mathcal{J}_\tau \coloneqq \{j \in \{0, \dots, \#\tau - 1\} : [\tau_j, \tau_{j+1}] \subseteq F\}$. These are the indices of the partition intervals on which $\mathbb{1}_F$ is identically 1. Therefore

$$L(\mathbb{1}_F;\tau) = \sum_{j \in \mathcal{J}_\tau} (\tau_{j+1} - \tau_j).$$

The set

$$K_\tau \coloneqq \bigcup_{j \in \mathcal{J}_\tau} [\tau_j, \tau_{j+1}]$$

is a compact subset of F. Consequently, it is covered by finitely many of the intervals $B_{r2^{-k}}(q_k)$. Applying the finite interval-cover inequality to each component of K_τ, and then summing over those components, gives

$$L(\mathbb{1}_F;\tau) \le \sum_{k=0}^{\infty} 2r2^{-k} = 4r.$$

Thus every upper sum equals 1, whereas every lower sum is at most $4r < 1$. Hence $\mathbb{1}_F$ is not Riemann integrable.

Step 3: rule out every Riemann-integrable limit.

Suppose that $g\colon [0,1] \to \mathbb{R}$ is Riemann integrable and $\|g - f_n\|_{L^1} \to 0$.
First we claim that

$$\int_0^1 g(x)\ \mathrm{d}x \le 4r < 1.$$

Indeed, from

$$\left| \int_0^1 g(x)\ \mathrm{d}x - \int_0^1 f_n(x)\ \mathrm{d}x \right| \le \|f_n - g\|_{L^1},$$

we obtain that

$$\int_0^1 g(x)\ \mathrm{d}x = \lim_{n\to\infty} \int_0^1 f_n(x)\ \mathrm{d}x.$$

Moreover,

$$0 \le f_n \le \sum_{k=0}^{n} \mathbb{1}_{B_{r2^{-k}}(q_k) \cap [0,1]},$$

and hence

$$\int_0^1 f_n(x)\ \mathrm{d}x \le \sum_{k=0}^{n} 2r2^{-k} \le 4r.$$

On the other hand, let $J \subseteq [0,1]$ be any nondegenerate interval. There is a rational number q_k in the interior of J, and hence a nondegenerate interval K of positive length $|K|$ such that

$$K \subseteq J \cap B_{r2^{-k}}(q_k).$$

For every $n \ge k$, we have $f_n = 1$ on K. If

$$\sup_{x \in J} g(x) < 1,$$

then, with

$$\eta \coloneqq 1 - \sup_{x \in J} g(x) > 0,$$

we would have, for every $n \geq k$,

$$\begin{aligned} d_1(f_n, g) &\geq \int_K |f_n(x) - g(x)| \, \mathrm{d}x \\ &= \int_K |1 - g(x)| \, \mathrm{d}x \\ &\geq \eta |K| > 0. \end{aligned}$$

This contradicts $d_1(f_n, g) \to 0$. Therefore

$$\sup_{x \in J} g(x) \geq 1$$

for every nondegenerate interval $J \subseteq [0, 1]$.
It follows that, for every partition τ,

$$U(g; \tau) \geq 1.$$

Since g is Riemann integrable,

$$\int_0^1 g(x) \, \mathrm{d}x = \inf_\tau U(g; \tau) \geq 1,$$

contradicting the inequality

$$\int_0^1 g(x) \, \mathrm{d}x < 1.$$

Thus (f_n) is Cauchy with respect to d_1, but it has no Riemann-integrable L^1-limit. Therefore the metric space obtained from the Riemann-integrable functions by identifying functions at L^1-distance zero is not complete. □